

In honour of Offer Kella · Markov-modulated AIMD

Exit Problems for AIMD in a Markovian Environment

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AIMD processes

AIMD means *Additive-Increase Multiplicative-Decrease*:
a positive process that grows gradually and is reduced proportionally at random decrease epochs.

additive increase

X grows linearly



random event

loss, congestion, reset



multiplicative decrease

$X \mapsto pX$

Pathwise intuition: upward crossings are continuous; downward crossings occur by jumps.

AIMD dynamics

Continuous-time model

$$\frac{dX_t}{dt} = \frac{1}{c}, \quad T_n \leq t < T_{n+1},$$

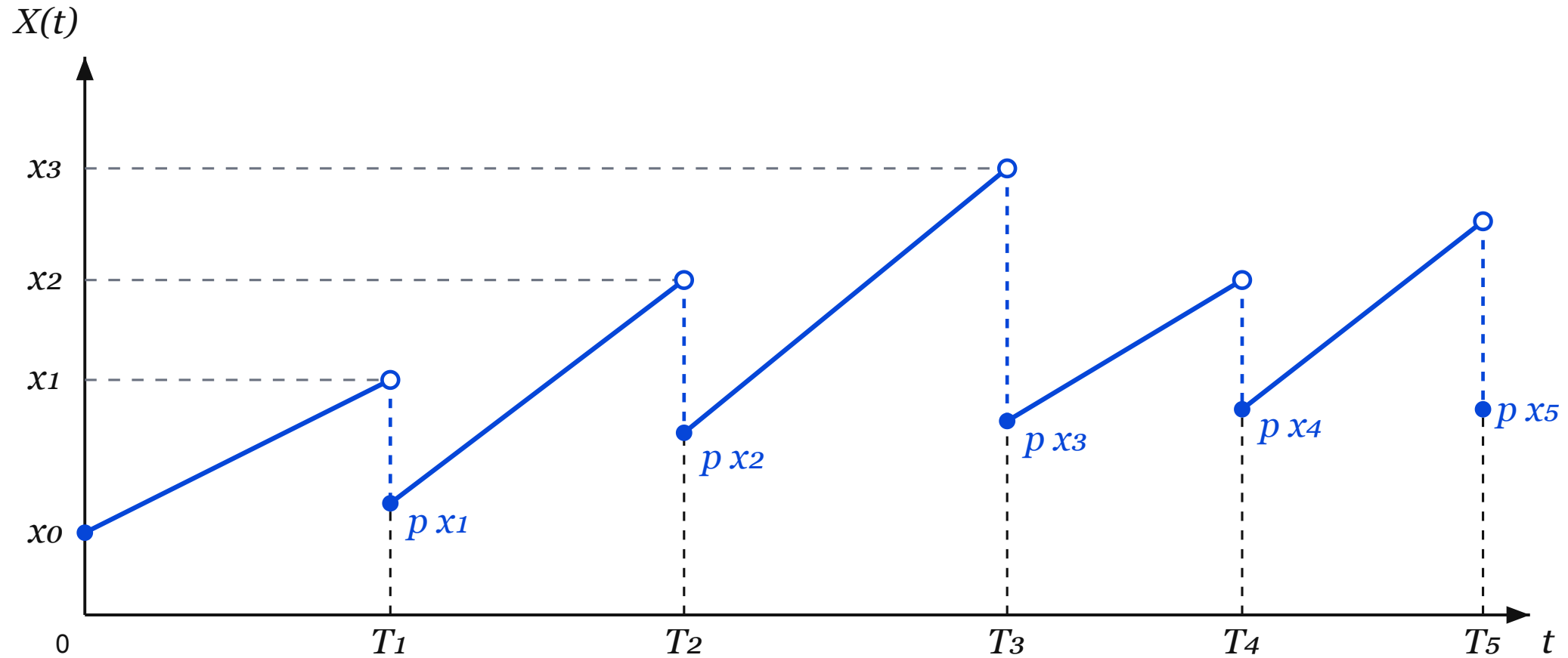
$$X(T_n) = pX(T_n-), \quad 0 < p < 1.$$

A **Poisson clock** with rate λ drives the decrease epochs.

Discrete-time model

$$X_{n+1} = \begin{cases} X_n + 1/c, & \text{with prob. } (1 - \delta), \\ pX_n, & \text{with prob. } \delta \end{cases} \quad 0 < p < 1.$$

AIMD sample path



Additive Increase

Between events, the process grows linearly with constant rate $a > 0$.



Multiplicative Decrease

At random epochs T_k , the process jumps to a fraction of its current value, that is, $X(T_k) = p_k X(T_k^-)$, $0 < p_k < 1$.

AIMD and TCP

TCP congestion control protocol is among classical applications:
it cautiously increases the *transmission rate* when transmission succeeds,
to decrease it sharply when congestion is detected.

Successful transmission

$$W \mapsto W + 1$$

Probe the available bandwidth by small additive steps.

Packet loss / congestion

$$W \mapsto pW, \quad p \approx \frac{1}{2}.$$

React quickly by reducing the current congestion window.

The same rule balances three goals: stability, fair bandwidth sharing, and adaptation to changing network conditions.

Earlier TCP/AIMD work

The TCP motivation led to a probabilistic literature on AIMD and growth-collapse dynamics.

Guillemin, Robert, Zwart (2004)

Ann. Appl. Probab. 14(1), 90-117.

AIMD algorithms and exponential functionals

Boxma, Perry, Stadje, Zacks (2006)

Adv. in Appl. Probab. 38(1), 221-243.

A Markovian growth-collapse model

van Leeuwaarden, Löpker, Ott (2009)

Queueing Syst. 63, 459-475.

TCP and iso-stationary transformations

AIMD and Partial resetting

AIMD is one instance of a broader class of **processes with partial resetting**: deterministic or stochastic growth interrupted by resets that keep part of the current state.

Search, restart and algorithms

Resetting appears in intermittent search and restart mechanisms, from abstract Markov processes to stress-release models.

[Evans et al. \(2020\)](#), [Evans and Majumdar \(2011\)](#),
[Gupta and Jayannavar \(2022\)](#), [Bénichou et al. \(2011\)](#),
[Avrachenkov et al. \(2013\)](#), [Borovkov and Vere-Jones \(2000\)](#).

Queueing, clearing and growth-collapse

Queueing applications include catastrophe rates, random clearing events and growth-collapse models.

[Artalejo et al. \(2006\)](#), [Kella et al. \(2003\)](#), [Boxma et al. \(2006\)](#),
[Löpker and Stadjé \(2011\)](#).

Our starting point:

2023 AIMD scalar model

The study [van der Hofstad et al. \(2023\)](#)[†] looked at exit-time identities for the scalar model.

The model is specified by one triplet (c, λ, p) :

- deterministic growth at rate $1/c$;
- Poisson jumps with rate λ ;
- mult. decrease factor $p \in (0, 1)$.

that determines the dynamics of the X process:

$$\begin{aligned}dX_t &= dt/c, & T_n \leq t < T_{n+1}, \\X(T_n) &= pX(T_n-), \\T_{n+1} - T_n &\sim \text{Exp}(\lambda).\end{aligned}$$

This talk - **Markovian modulation**

A finite-state Markov environment J modulates all local parameters.

$$(c_i, \lambda_i, p_i), \quad Q = (q_{ij}), \quad i, j \in \{1, \dots, m\}.$$

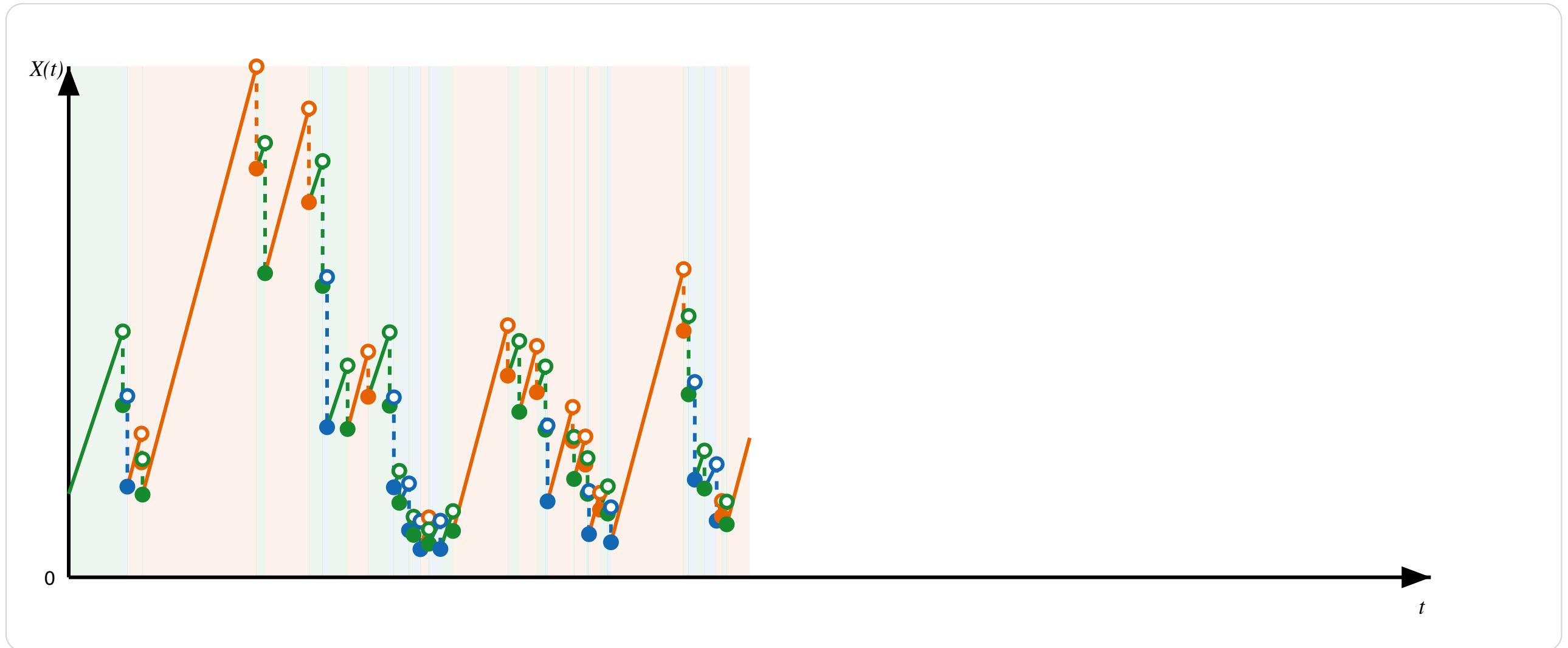
The scalar transforms of the exit-times become **matrix-valued** transforms, indexed by the starting and ending environment regime.

$$\begin{aligned}\{J(T_n) = i\} &\implies \{J(T_{n+1}) = j\} \text{ with prob. } q_{ij}, \\dX_t &= dt/c_i, \quad J(t) = i & T_n \leq t < T_{n+1} \\X(T_n) &= p_i X(T_n-), \\T_{n+1} - T_n &\sim \text{Exp}(\lambda_i).\end{aligned}$$

Note: the skip-free/first-step strategy survives, but scalar q -series are replaced by non-commutative regime-indexed matrix series.

[†]R. van der Hofstad, S. Kapodistria, Z. Palmowski, S. Shneer. J. Appl. Probab. 60 (2023)

Markov modulated AIMD simulating one sample path



☒ Hide future times



Objects of interest

For fixed levels $0 \leq b < a$, define

$$\tau_a^\uparrow = \inf\{t \geq 0 : X_t \geq a\}, \quad \tau_b^\downarrow = \inf\{t \geq 0 : X_t \leq b\}.$$

Upward one-sided transform $Z(x \uparrow a)$

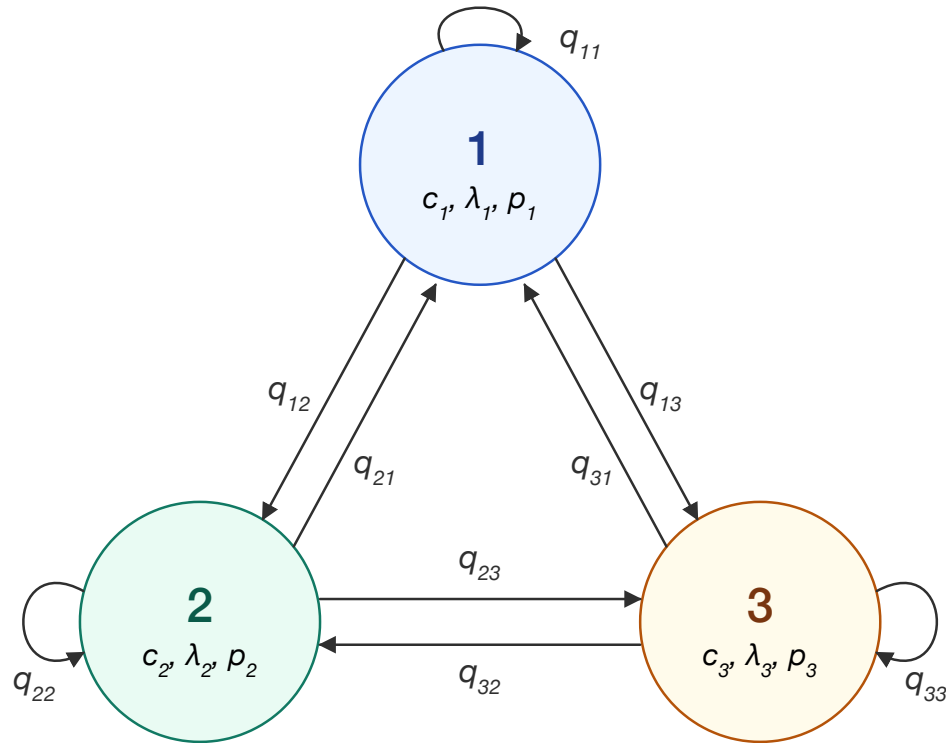
$$Z_{ij}(w, x \uparrow a) = \mathbb{E}_{x,i}[e^{-w\tau_a^\uparrow}; J(\tau_a^\uparrow) = j].$$

Two-sided upward transform $L(x \uparrow_b^a)$

$$L_{ij}(w, x \uparrow_b^a) = \mathbb{E}_{x,i}[e^{-w\tau_a^\uparrow}; \tau_a^\uparrow < \tau_b^\downarrow, J(\tau_a^\uparrow) = j].$$

Throughout, matrix products encode conditioning on the regime reached at an intermediate level.

A closer look at the introduction of the Markovian environment



Starting after a jump at time t_n at level x_n and state i_n , the level just after a new jump at time t_{n+1} is

$$x_{n+1} = p_{i_n} \left(x_n + \frac{t_{n+1} - t_n}{c_{i_n}} \right).$$

The next state i_{n+1} is selected randomly according to the distribution given by the i -th row of transition matrix Q .

Hence a first-step equation mixes:

- state-dependent linear growth $t \mapsto x + t/c_i$
- state-dependent rescaling $x \mapsto x/p_i$
- multiplication by selectors E_{ii}
- matrix products with Q .

Some notation

Let $\mathbf{z} = (z_i)$ and $\boldsymbol{\gamma} = (\gamma_i)$, with:

$$z_i = c_i(\lambda_i + w), \quad \gamma_i = \frac{c_i \lambda_i}{p_i}.$$

Diagonal and selector matrices:

$$\begin{aligned} \boldsymbol{\mathcal{E}}(\mathbf{v}) &= \mathbf{diag}(e^{v_i}), & \boldsymbol{\Delta}(\mathbf{v}) &= \mathbf{diag}(v_i), \\ \mathbf{E}_i &= (\delta_{ik}\delta_{il})_{kl}. \end{aligned}$$

Some useful matrices:

$$\mathbf{A}(s) = \boldsymbol{\Delta}\left(\frac{\boldsymbol{\gamma}}{\mathbf{z} - s}\right), \quad \mathbf{B} = -\boldsymbol{\Delta}(\boldsymbol{\gamma})^{-1}.$$

Upward one-sided exit: $\mathbf{Z}(x \uparrow a) = \mathbb{E}_x[e^{-w\tau_a^\uparrow}; J(\tau_a^\uparrow)]$

For $0 < x < a$, upward passage is continuous, so the process must cross x before a .

$$\mathbf{Z}(0 \uparrow a) = \mathbf{Z}(0 \uparrow x) \mathbf{Z}(x \uparrow a).$$

Consequently,

$$\boxed{\mathbf{Z}(x \uparrow a) = \mathbf{Z}(0 \uparrow x)^{-1} \mathbf{Z}(0 \uparrow a)}$$

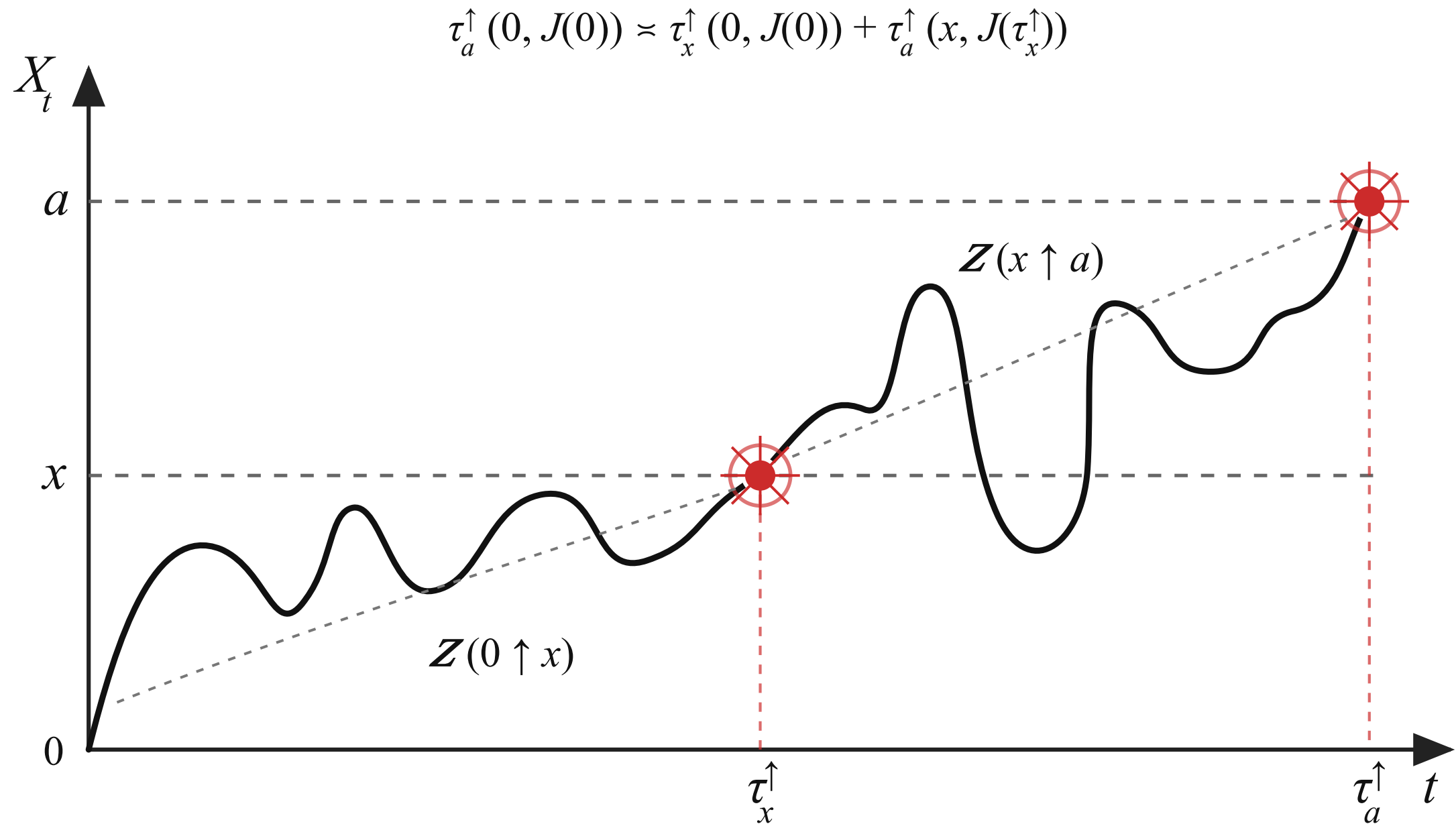
and we are left with computing $\mathbf{Z}(0 \uparrow x)$ for any $x \geq 0$.

The proof is the strong Markov property at $\tau_x^\uparrow$, plus local invertibility of the matrix evolution.

Main ingredient

$$\tau_a^\uparrow(0, J(0)) \stackrel{\text{dist}}{=} \tau_x^\uparrow(0, J(0)) + \tau_a^\uparrow(x, J(\tau_x^\uparrow(0, J(0)))).$$

Strong Markov property at the hitting times



Computing the upward scale matrix

First-step analysis at the first jump gives

$$\mathbf{I} = \mathcal{E}(-a\mathbf{z})\mathbf{Z}^{-1}(0 \uparrow a) + \Delta(\gamma) \int_0^\infty \Delta(x \leq a\mathbf{p}) \mathcal{E}(-(z/\mathbf{p})x) \mathbf{Q} \mathbf{Z}^{-1}(0 \uparrow x) dx.$$

split at first jump

$\Rightarrow$

Volterra-type matrix equation

$\Rightarrow$

Laplace functional equation

Here $(x \leq a\mathbf{p})_i = \mathbb{1}\{x \leq ap_i\}$.

Upward one-sided exit: Laplace-domain solution

Let $\tilde{U}(s) = \int_0^\infty e^{-sa} \mathbf{Z}^{-1}(0 \uparrow a) da$. Then

$$\tilde{U}(s) = \mathbf{A}(s)\mathbf{B} + \mathbf{A}(s) \sum_{i=1}^m \mathbf{E}_i \mathbf{Q} \tilde{U}(s/p_i).$$

Iterating this equation yields

$$\tilde{U}(s) = \mathbf{A}(s)\mathbf{D}(s)\mathbf{B}$$

$$\mathbf{D}(s) = \mathbf{I} + \sum_{n \geq 1} \sum_{i_1, \dots, i_n} \mathbf{E}_{i_1} \mathbf{Q} \mathbf{A}(s/p_{i_1}) \cdots \mathbf{E}_{i_n} \mathbf{Q} \mathbf{A}\left(\frac{s}{p_{i_1} \cdots p_{i_n}}\right).$$

The series converges because repeated multiplications by $p_i < 1$ push the residuals quickly towards 0.

Comparison with the scalar benchmark

Scalar AIMD

$$Z^\uparrow(w; x, a) = \frac{Z^\uparrow(w; 0, a)}{Z^\uparrow(w; 0, x)}$$

$$Z^\uparrow(w; 0, x) = \left(\sum_{k=0}^{\infty} \frac{(x/(w + \lambda))^k}{k!} (\lambda/(w + \lambda); p)_k \right)^{-1}.$$

Markov-modulated AIMD

Division is replaced by matrix inversion.

A scalar q -series is replaced by the non-commutative series $\mathbf{D}(s)$.

The matrices $\mathbf{E}_i \mathbf{Q}$ trace the regimes selected at each jump.

For $m = 1$, the matrix formula collapses to the scalar recursion from the previous paper.

From q -calculus: $(x; q)_n = (1 - x)(1 - xq)(1 - xq^2) \cdots (1 - xq^{n-1})$

The downward correction starts at the boundary

Write $\mathbf{Z}_b(x) = \mathbf{Z}(x \downarrow b)$, with $\mathbf{Z}_b(x) = \mathbf{I}$ for $x \leq b$.

The first-step equation is valid only for $x > b$:

$$\vec{e}_i \mathbf{Z}_b(x) = \gamma_i e^{z_i x} \int_{p_i x}^{\infty} \vec{e}_i \mathbf{Q} \mathbf{Z}_b(y) e^{-(z_i/p_i)y} dy.$$

Incorrect domain

$$\int_0^{\infty} e^{-sx} \mathbf{Z}_b(x) dx$$

extends the renewal equation below b , where it does not hold.

Correct spatial transform

$$\tilde{\mathbf{Z}}_b(s) = \int_b^{\infty} e^{-sx} \mathbf{Z}_b(x) dx$$

The known interval $[p_i b, b]$ survives as an explicit forcing term.

The same Green operator now carries boundary forcing

For the known diagonal matrix $\mathbf{R}_b(s) = \text{diag}(R_i(s; b))$, set

$$\mathbf{C}_b(s) = \mathbf{R}_b(s)\mathbf{Q} + e^{-bs}\mathbf{H}_b, \quad \mathbf{H}_b = -\sum_{i=1}^m e^{z_i b} \mathbf{E}_i \mathbf{Q} \tilde{\mathbf{Z}}_b(z_i/p_i).$$

$$R_i(s; b) = \frac{p_i}{s} (e^{-sb} - e^{-sb/p_i}) - \frac{p_i}{z_i} e^{(z_i-s)b} (e^{-z_i b} - e^{-z_i b/p_i}).$$

The regime-word Green operator is

$$(\mathcal{D}F)(s) = F(s) + \sum_{n \geq 1} \sum_{|\alpha|=n} \mathbf{K}_\alpha(s) F(s/p_\alpha).$$

Therefore

$$\tilde{\mathbf{Z}}_b(s) = \mathbf{A}(s)(\mathcal{D}\mathbf{C}_b)(s).$$

What stays the same?

The word kernels $\mathbf{K}_\alpha$.

What changes?

The forcing remembers b .

What remains unknown?

Only the finite matrix $\mathbf{H}_b$, obtained from an $m \times m$ consistency system.

Two-sided upward exit: $\mathbf{L}(x \uparrow_b^a) = \mathbb{E}_x[e^{-w\tau_a^\uparrow}; \tau_a^\uparrow < \tau_b^\downarrow, J(\tau_a^\uparrow)]$

Let $\mathbf{L}(x \uparrow_b^a)$ denote the transform of the upward exit time before downward exit.
We use a constructive recursion to compute it.

Assuming $\mathbf{L}(y \uparrow_b^a)$ known for all $y < a$, with $\rho < \min_i p_i$,
then $\mathbf{L}(x \uparrow_b^{a/\rho})$ is obtained for $x = a$ as

$$\mathbf{L}(a \uparrow_b^{a/\rho}) = (I - \mathbf{H}(a))^{-1} \mathcal{E}(-(a/\rho - a)\mathbf{z}) \mathbf{L}(a \uparrow_b^a),$$

and for $x < a/\rho$ as

$$\mathbf{L}(x \uparrow_b^{a/\rho}) = \mathcal{E}(-(a/\rho - x)\mathbf{z}) + \mathbf{H}(x) \mathbf{L}(a \uparrow_b^{a/\rho}).$$

The matrix $\mathbf{H}(x)$ with components $H_{ij}(x) = \gamma_i e^{z_i x / p_i} \int_{p_i x}^{p_i a / \rho} \sum_h q_{ih} L_{hj}(w, y \uparrow_b^a) e^{-(z_i / p_i) y} dy$ can be explicitly computed.

Reflection requires remembering the current minimum

Start with an initial minimum $u > 0$ and define

$$V_t = u \wedge \inf_{0 \leq s \leq t} X(s), \quad \tilde{Y}_t = X(t) - V_t, \quad \tilde{\tau}_c = \inf\{t \geq 0 : \tilde{Y}_t \geq c\}.$$

State that is Markov

Keep (X_t, V_t, J_t) .

A downward jump may create a new minimum, so the continuation law depends on its absolute landing level and phase.

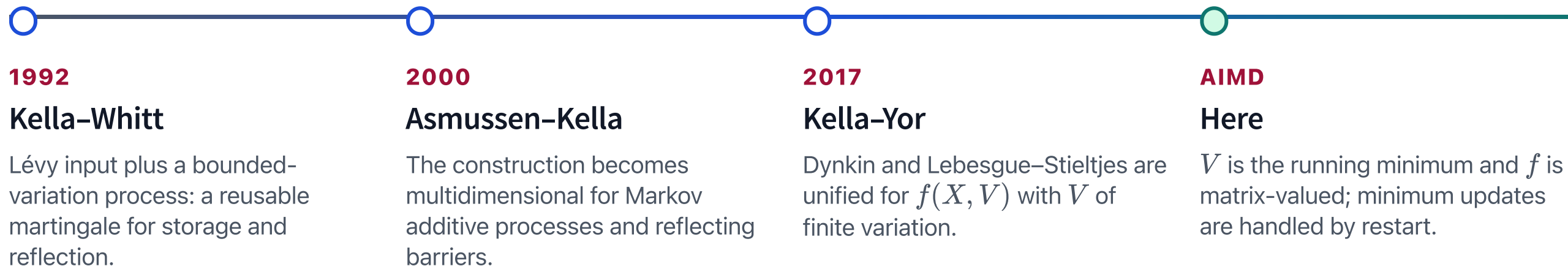
State that is not enough

$\tilde{Y}_t$ alone forgets where the next multiplicative jump lands.

Fixed-barrier scale functions alone do not retain the information needed after a new minimum.

The diagonal state (v, v) is regenerative, but its continuation value is a matrix function of v .

A martingale line running through Kella's work



Hand-off from the preceding talk: the Markov phase suggests Asmussen–Kella; the moving minimum calls for the Kella–Yor finite-variation viewpoint.

Restart at the next strict minimum

At a current minimum v , let

$$\rho_v = \inf\{t > 0 : X(t) < v\}.$$

Successful excursion

$$S_{ij}(v) = \mathbb{E}_{v,i} \left[e^{-w\tau_{v+c}^\uparrow}; \tau_{v+c}^\uparrow < \rho_v, J(\tau_{v+c}^\uparrow) = j \right].$$

Restart operator

$$(TG)_{ij}(v) = \mathbb{E}_{v,i} \left[e^{-w\rho_v} G_{J(\rho_v),j}(X(\rho_v)); \rho_v < \tau_{v+c}^\uparrow \right].$$

The continuation matrix at the diagonal is the unique bounded solution

$$\mathbf{H} = \mathbf{S} + \mathbf{T}\mathbf{H}, \quad \mathbf{H} = \sum_{n \geq 0} \mathbf{T}^n \mathbf{S}.$$

Indeed $\|\mathbf{T}\| \leq \max_i \frac{\lambda_i}{w+\lambda_i} (1 - e^{-(w+\lambda_i)c_i c}) < 1$.

The continuation function is harmonic between updates

For $v \leq x \leq v + c$, define

$$F_{ij}(w; x, v) = \mathbb{E}_{x,i} \left[e^{-w\tau_{v+c}^\uparrow}; \tau_{v+c}^\uparrow < \rho_v, J(\tau_{v+c}^\uparrow) = j \right] \\ + \mathbb{E}_{x,i} \left[e^{-w\rho_v} H_{J(\rho_v),j}(X(\rho_v)); \rho_v < \tau_{v+c}^\uparrow \right].$$

$$F(w; v, v) = \mathbf{H}(v), \quad F(w; v + c, v) = \mathbf{I}.$$

Extend $F(x, v) = \mathbf{H}(x)$ for $x \leq v$.

First-step conditioning gives

$$\mathcal{A}^\dagger F(\cdot, v)(x) = 0, \quad v < x < v + c.$$

The lower branch automatically selects the new-minimum value after an overshoot.

Kella–Yor turns restart harmonicity into a martingale

Stopped at $\tilde{\tau}_c$,

$$M_t = e^{-w(t \wedge \tilde{\tau}_c)} \vec{e}_{J(t \wedge \tilde{\tau}_c)} F(X(t \wedge \tilde{\tau}_c), V_{t \wedge \tilde{\tau}_c})$$

is a bounded martingale.

Between jumps

V is fixed and $\mathcal{A}^\dagger F = 0$ cancels the deterministic and discount drifts.

At a new minimum

The jump lands on the diagonal and $F(v, v) = H(v) = S(v) + (TH)(v)$ supplies the correct restart value.

Optional stopping

$0 \leq F\mathbf{1} \leq \mathbf{1}$, so the local martingale is bounded and hence a true martingale.

The reflected first-passage transform follows

For $u \leq x < u + c$, define

$$R_{ij}(w; x, u; c) = \mathbb{E}_{x,i} \left[e^{-w\tilde{\tau}_c}; J(\tilde{\tau}_c) = j \right].$$

$$\mathbf{R}(w; x, u; c) = \mathbf{F}(w; x, u), \quad \mathbf{R}(w; u, u; c) = \mathbf{H}(u).$$

If $x - u \geq c$, then $\mathbf{R}(w; x, u; c) = \mathbf{I}$.

Spatial Green operator $\mathcal{D}$

Sums ordinary AIMD jump words in the Laplace variable.

Used by the corrected downward transform.

Restart Green series $\sum_{n \geq 0} T^n$

Sums complete passages from one strict running minimum to the next.

Used by the reflected process.

They are analogous, not interchangeable: restart retains the overshoot level; the downward transform does not.

Take-home points

1

Upward and downward passage use the same regime-word Green kernel, but the corrected downward forcing must remember the boundary b .

2

Reflection is a restart problem: the continuation matrix $H(v)$ records the overshoot level at every strict new minimum.

3

The Kella–Yor martingale converts restart harmonicity into the full matrix transform of the reflected first-passage time.

Questions

Thank you.

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Joint work with Zbigniew Palmowski

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